

Module 3: Data Warehousing & OLAP Cubes

Star/Snowflake Schemas, Data Cube Computation & OLAP Operations

Module III: Data Warehousing & OLAP Cubes — Topics Covered

1. OLTP vs. OLAP Multidimensional Modeling
2. Schemas: Star, Snowflake & Fact Constellation
3. Lattice of Cuboids & Data Cube Computation
4. OLAP Operations: Roll-up, Drill-down, Slice, Dice, Pivot
5. Attribute-Oriented Induction (AOI)

1. Core OLAP Operations

- **Roll-Up (Drill-Up):** Summarizes data by climbing up concept hierarchies (e.g., Day → Month → Quarter → Year).
- **Drill-Down (Roll-Down):** Navigates from higher-level summary to detailed granular data.
- **Slice:** Performs a selection on one dimension (e.g., `Time = "Q1"`).
- **Dice:** Defines a sub-cube by selecting on two or more dimensions (e.g., `Time = "Q1"` and `Location = "India"`).
- **Pivot (Rotate):** Rotates multidimensional axes to provide an alternative perspective.